JAGRUTHI GUTTULA

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GitHub | LinkedIn | Portfolio

EDUCATION

Gokaraju Lailavathi Engineering College | B.Tech, Computer Science Engineering | Expected 2029

SELECTED PROJECTS

KIKO - Accessible Desktop Learning Assistant | Python, Windows, Ollama (exploratory)
Prototyped a beginner-friendly assistant that walks users through computer tasks; built a guided Practice Lab with step-by-step help and progress tracking. Developing multilingual and offline voice guidance.

TREE - Home Energy Retrofit Scoring | Python, sustainability scoring, data analysis
Built a Green Home Score framework evaluating household energy efficiency, insulation, heating, glazing, renewables, and smart features to explore retrofit decisions; advanced to Round 2 at IIITH MegathonX 2026.

ASTRA - Spacecraft Fault-Recovery Testbed | Dual ESP32, DS18B20, OLED, FastAPI
Prototyped fault-response states across two ESP32 boards, modeling anomaly detection, payload isolation, and recovery; integrating temperature sensing, status feedback, and a monitoring dashboard.

Interactive Smart Nerf Controller | ESP32, MPU6050, piezo sensor, OLED, Arduino/C++
Built a motion-aware physical controller combining impact detection, buttons, LEDs, buzzer, and OLED feedback; developed in a two-person team for Hack'O'Ber 2026 (2nd place).

RaceOS - F1 Pit-Stop Strategy Assistant | Python, FastF1, scikit-learn, Streamlit
Engineered Formula 1 lap telemetry features and trained a rebalanced Random Forest pipeline to predict next-lap pit stops (~85.9% reported test accuracy); built an interactive Streamlit interface.

ChronoMind AI - Personalized Focus & Scheduling | Python, Pandas, scikit-learn, Next.js
Investigated cognitive rhythms and focus windows using 3,106 observations; compared population and personalized baselines (MAE 77.4 vs. 45.8) and prototyped cognitive games and scheduling views.

Trekker - Safety Wearable | ESP32, GNSS, MAX30102, MPU6050, BLE
Prototyped a safety wearable combining location, pulse/SpO2, motion sensing, and physical SOS input; designed offline-first alerts and geofencing workflows for low-connectivity environments.

CURRENTLY BUILDING

Circuit Studio - Interactive Electronics Playground | ESP32, I2C, Web Serial, visual rule engine
Building an experimentation-first electronics studio where beginners discover connected components, follow breadboard wiring guides, and create visual input-to-condition-to-output rules for LEDs and buzzers; physical integration is in progress.

PARADOX - Museum of the Unbelievable | Interactive web, perception, optical illusions
Developing a digital museum with hands-on optical-illusion exhibits exploring perspective, mirrors, impossible spaces, and color perception; exploring VR as a future extension.

Roadster - Miniature Tesla-Inspired Vehicle | RC vehicle, embedded systems, sensors
Designing a miniature electric-vehicle prototype to explore motor control, onboard sensing, and hardware-software integration.

MILO - Mobile Elderly-Care Companion | RC chassis, ESP32, computer vision, wearable sensing
Developing a mobile teddy-bear companion with voice interaction, camera-assisted check-ins, and a wearable heart-rate, SpO2, motion, and SOS band to support caregiver alerts.

EXPERIENCE

SaimlSol Private Limited - AI/ML Intern
Supported Python-based AI/ML and IoT development, data preparation, and model experimentation.

AWARDS & LEADERSHIP

Hack'O'Ber 2026 - 2nd place (two-person team) | IIITH MegathonX 2026 - Advanced to Round 2 with TREE.
Dexterity School of Leadership & Entrepreneurship - Selected among 100 young leaders globally.
NGFP Young Voices - Global Rank #30 | RoundPier - Junior Fellow and Best Presentation Award.

TECHNICAL SKILLS

Languages / ML: Python, C, C++, C#, TypeScript, SQL; scikit-learn, Pandas, NumPy, OpenCV, YOLO
Web / Embedded: React, Next.js, FastAPI, Streamlit, PostgreSQL, Git/GitHub; ESP32, Arduino, I2C, sensors